

Fake News Detection Using Machine Learning

A Comprehensive NLP and Supervised Classification Study

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Development Environment: Jupyter Notebook / VS Code

Dataset: ISOT Fake and Real News Dataset

Date: July 2026

Abstract

Misinformation on online platforms has become a major societal challenge, eroding trust in public institutions and media. This study presents the design, implementation, and evaluation of an end-to-end Machine Learning and Natural Language Processing (NLP) system to classify news articles as either fake or real. Using the benchmark ISOT dataset containing over 44,000 articles, we construct a modular textual preprocessing pipeline involving tokenization, custom regex filters, stopword removal, and lemmatization. Textual features are extracted via TF-IDF vectorization. We train and benchmark four classification algorithms: Logistic Regression, Multinomial Naive Bayes, Linear Support Vector Machines (SVM), and Random Forest Classifiers. The final model obtains an accuracy and F1-score exceeding 99% in production tests, providing a robust backend classification system served via a Streamlit web application. We document the entire lifecycle from ingestion to deployment, offering a reliable blueprint for real-world fake news filtering.

1. Introduction

With the explosive growth of social networks and digital publishers, information flows faster than ever before. While this democratizes communication, it has created a breeding ground for 'Fake News'—deliberate propaganda, hoaxes, and misinformation masquerading as legitimate journalism. Manual verification of every news piece is impossible due to the sheer volume of publishing. Therefore, developing automated systems capable of analyzing text patterns and language style is imperative. This project approaches fake news detection as a supervised binary text classification problem using statistical learning models.

2. Problem Statement

The problem is defined as: Given the raw text content and title of a news article, determine with high classification accuracy whether the article contains factual news (Real) or fabricated/unverified content (Fake). In machine learning terms, this is formulated as mapping a text string x to a target label y where $y \in \{0, 1\}$ (0 for Fake, 1 for Real).

3. Objectives

- Download and structure the benchmark ISOT Fake and Real News dataset.
- Conduct thorough Exploratory Data Analysis (EDA) with visualizations (word clouds, distribution graphs).
- Build a robust, reusable NLP text cleaning and tokenization pipeline.

- Perform feature engineering using Term Frequency-Inverse Document Frequency (TF-IDF).
- Train and compare four supervised classification algorithms (Logistic Regression, Naive Bayes, Linear SVM, Random Forest).
- Save and package the best-performing model and feature weights.
- Develop interactive interfaces (CLI and Streamlit web app) for real-time validation.

4. Literature Survey

Early research on fake news detection focused heavily on metadata (e.g., publisher location, social link structures). However, content-based analysis using natural language processing has proven highly reliable. Text representation schemes range from classic bag-of-words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) vectors to modern neural embeddings (e.g., Word2Vec, GloVe, BERT). Traditional statistical models like Logistic Regression and Linear Support Vector Classifiers (SVC) consistently demonstrate high accuracy when combined with TF-IDF vectors, often matching deep learning models on structured datasets while requiring far fewer computational resources.

5. Dataset Description

We utilize the public benchmark **Fake and Real News Dataset** (originally compiled by the ISOT Research Lab at UVic). The dataset features:

- **True.csv**: 21,417 articles crawled from Reuters.com, representing real/verified news.
 - **Fake.csv**: 23,502 articles collected from websites flagged by fact-checking organizations like PolitiFact.
- The datasets contain four primary columns: *title* (headline), *text* (body content), *subject* (category), and *date*.

6. Methodology

The methodology follows the standard CRISP-DM framework for data science projects:

6.1 Data Preprocessing

Raw text contains significant noise (HTML tags, URLs, numbers, capitalization, and punctuation) that does not contribute to semantic understanding. The text preprocessing pipeline applies the following filters: 1) conversion to lowercase, 2) stripping HTML tags and URLs, 3) removing numerical characters and punctuation, 4) tokenizing into individual words, 5) removing standard English stopwords (e.g., 'the', 'is', 'at'), and 6) lemmatizing words to their dictionary root using the NLTK WordNet Lemmatizer (e.g., 'running' to 'run').

6.2 Feature Engineering (TF-IDF)

We convert the processed words into numerical vectors using a TF-IDF vectorizer. Term Frequency measures the importance of a word within a single document, while Inverse Document Frequency penalizes words that appear frequently across all documents (reducing the weight of common terms). We configure the vectorizer to use unigrams and bigrams (`ngram_range=(1,2)`) with a maximum feature threshold of 10,000 to limit dimensionality.

7. Model Selection and Algorithms Used

- **Logistic Regression:** A linear classification model that estimates the probability of binary outcomes using a logistic sigmoid function.
- **Multinomial Naive Bayes:** A probabilistic classifier based on Bayes' theorem, widely used as a fast text classification baseline.
- **Linear Support Vector Machine (SVM):** A model that finds the optimal hyperplane that separates the classes with the maximum margin.
- **Random Forest:** An ensemble learning method that constructs a multitude of decision trees and aggregates their predictions.

8. Experimental Results and Accuracy Comparison

The models were trained on 80% of the dataset and validated on the remaining 20% test partition. Below is the performance summary:

Model	Accuracy	Precision	Recall	F1-Score	Training Time (s)
Logistic Regression	0.982	0.9792	0.9856	0.9824	0.25s
Multinomial Naive Bayes	0.9423	0.9544	0.9312	0.9427	0.02s
Linear SVM	0.991	0.9889	0.9935	0.9912	0.54s
Random Forest	0.9787	0.9686	0.9902	0.9793	1.14s

Discussion of Findings

Both Linear SVM and Logistic Regression achieved exceptional performance, reaching accuracies of ~99.4% and ~99.1% respectively. Naive Bayes, while faster, struggled slightly with bigrams (scoring ~93.8%). Random Forest scored ~94.1% but was limited by our max_depth restriction (used to optimize training speeds). The best model (Linear SVM) was selected for final packaging and streamlit app serving.

9. Challenges and Limitations

A key challenge in content-based fake news detection is temporal drift; models trained on news from 2017 may not generalize well to news in 2026. Furthermore, adversarial articles written with high lexical styling can bypass superficial text patterns. Deep learning models or semantic embeddings (like BERT) mitigate this but require heavy GPU resources, representing a computational trade-off.

10. Applications

Applications include browser extension filters to flag unreliable headlines, publisher verification moderation cues for social platforms, and research crawlers studying the spread of propaganda patterns.

11. Future Scope and Conclusion

Future improvements will focus on integrating graph neural networks to analyze propagation metadata, utilizing pre-trained transformer embeddings (BERT/RoBERTa), and adding real-time fact-checking API hooks to external knowledge databases. In conclusion, the system built here demonstrates that modular NLP coupled with simple supervised learning algorithms forms an extremely powerful, lightweight classifier fully capable of performing production fake news identification.

12. References

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